

FATmax Studio Pro (Group Edition)

Official User Manual & Technical Guide

1. Introduction

FATmax Studio Pro is a fully automated, web-based analytical platform designed for sports scientists, physiologists, and clinicians. It facilitates the batch processing of up to 30 subjects simultaneously, extracting key fat oxidation kinetics—such as Maximal Fat Oxidation (MFO), FATmax, and FATmin—using advanced mathematical modeling. This group edition dramatically accelerates data analysis by replacing manual, subject-by-subject curve fitting with an instantaneous, automated cohort pipeline. It outputs comprehensive visual grids, group cohort averages, and publication-ready statistical datasets.

2. App Engine & Technical Innovations

- **Advanced CPET-Specific Imputation:** Traditional rolling means often blunt or erase the true physiological peaks in short-stage CPET data. FATmax Studio Pro utilizes a 2nd-degree polynomial baseline paired with a 2.5 Standard Deviation threshold to identify true outliers (e.g., erratic breathing). Gaps are then corrected using safe interior linear interpolation, preserving the natural boundaries of the curve.
- **Dual-Model Engine:** The app simultaneously computes an Exponential Sine model alongside an innovative 'Resting-In' model. This dual approach provides a comprehensive view of both the isolated exercise response and the total physiological response.
- **Wald's Runs Test for Randomness:** Pure variance metrics like R^2 can be misleading, as a model might have a high R^2 but systematically overestimate early stages. The engine employs Wald's Runs Test to detect structural non-randomness in the residuals. If $p < 0.05$, it flags the model as a 'Poor Fit' due to unnatural structural deviation, providing a highly robust clinical safety net.

3. Mathematical Models

While previous methodologies (e.g., Cheneviere et al., 2009) utilized bounded formulas restricted to relative percentages (%VO2max), FATmax Studio Pro utilizes a dynamically scaling **Exponential Sine Equation**. This allows the model to process raw absolute workloads natively, providing highly stable computational fitting without numerical domain errors or relying on secondary root-finding algorithms.

Model 1: The Exponential Sine Model (0,0 Intercept)

$$y = a * \sin(\pi * x / b) * \exp(-c * x)$$

Mathematically, the sine function forces the curve to anchor exactly at (0,0). While humans biologically oxidize fat at rest, this model is specifically designed to isolate and map the exercise-induced kinetic response without resting noise.

- **Amplitude (a):** Functions as a vertical multiplier scaling the curve to the subject's exact peak fat oxidation rate (MFO) without altering the starting coordinates at load=0.
- **Dilatation (b):** Establishes the horizontal period of the wave. It determines exactly where the sine function naturally returns to zero, mapping the true mathematical inhibition of fat oxidation (FATmin).
- **Symmetry (c):** An exponential decay factor. Positive values dynamically decay the wave to shift the peak leftwards (early activation), while negative values amplify the wave, pushing the peak rightwards (delayed inhibition). No rigid horizontal translation is utilized to preserve the baseline anchor.

Model 2: The 'Resting-In' Model (+d Baseline)

$$y = a * \sin(\pi * x / b) * \exp(-c * x) + d$$

This innovation introduces a baseline constant (+d). This allows the curve to anchor at the athlete's true resting oxidation rate, preventing the mathematical distortion of amplitude and symmetry when analyzing sedentary, obese, or diabetic cohorts with high resting fat oxidation.

Clinical Insight on Dual Modeling: Incorporating the resting value (+d) typically does not drastically alter the absolute values of MFO, FATmax, or the FATmax Zone. However, it significantly alters the structural shape parameters (Symmetry, Amplitude, and Dilatation). Comparing both models allows clinicians to observe how resting metabolism impacts the mathematical curve structure without compromising peak thresholds.

Contains a side-by-side analysis of both models for every subject:

- **MFO [g/min]:** Maximal Fat Oxidation (Measured and Modeled).
- **MFO Bias:** The exact difference (Modeled - Measured) to track over/under-estimation.
- **FATmax & FATmin:** The load at which oxidation peaks, and the load where it crosses back to zero.
- **FATmax Zone:** The specific load bandwidth where fat oxidation is $\geq 90\%$ of MFO.
- **Shape Parameters:** Dilatation (b), Symmetry (c), and Resting Baseline (d).
- **Quality Metrics:** Outliers Imputed, R^2 (Fit Variance), and Wald's Runs Test p-value.

Sheet 2: Model Concordance (CCC)

This sheet mathematically compares the Original and Resting-In models across the entire cohort using Lin's Concordance Correlation Coefficient (CCC). This allows researchers to quickly assess agreement and determine if the resting baseline significantly impacts specific parameters like Symmetry or Dilatation.

Sheet 3: Raw Input Data

A direct copy of your original .csv dataset for complete transparency and replicability.